EXERCISE 3

$D = \begin{cases} 1 & \text{person has the condition} \\ 0 & \text{otherwise} \end{cases}$

$T = \begin{cases} 1 & \text{person is tested positive} \\ 0 & \text{otherwise} \end{cases}$

$P_{T|D}(1|1) = 0.95$ $\qquad$ $P_{T|D}(1|0) = 0.01$

$\downarrow P(T=1|D=1)$ $\qquad$ $\downarrow P(T=1|D=0)$

$P_{T|D}(0|1) = 0.05$ $\qquad$ $P_{T|D}(0|0) = 0.99$

$P_D(1) = P(D=1) = 0.005$

$$P(D=1|T=1) = P_{D|T}(1|1) = \frac{P_{D,T}(1,1)}{P_T(1)} = \frac{\overset{0.005}{P_D(1)}\,\overset{0.95}{P_{T|D}(1|1)}}{?= P_T(1)} \; *$$

$$P_T(1) = P_{T,D}(1,1) + P_{T,D}(1,0) =$$

$$= \boxed{P_D(1)\,P_{T|D}(1|1)} + \underset{0.995}{P_D(0)}\;\underset{0.01}{P_{T|D}(1|0)}$$

$$= \frac{0.00475}{0.00475 + 0.0095} = 0.3231$$